SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 5

SMART: Simple Multi-Attribute Rating Technique

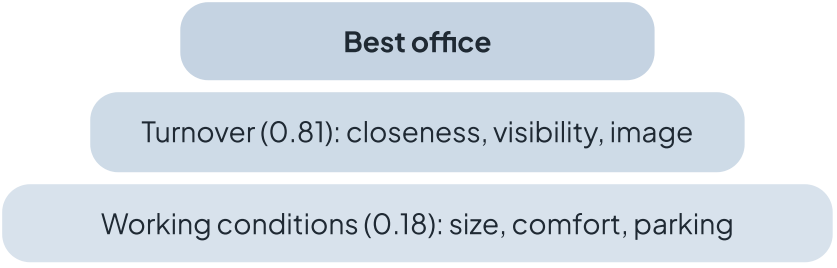
SMART for multi-criteria choices: value tree, 0–100 values, swing weights, weighted aggregate, cost vs benefit (efficient frontier), sensitivity analysis. Class example: Bandung office location. Your syndicate: BUMDes Bumi Lestari's bank selection.

Material

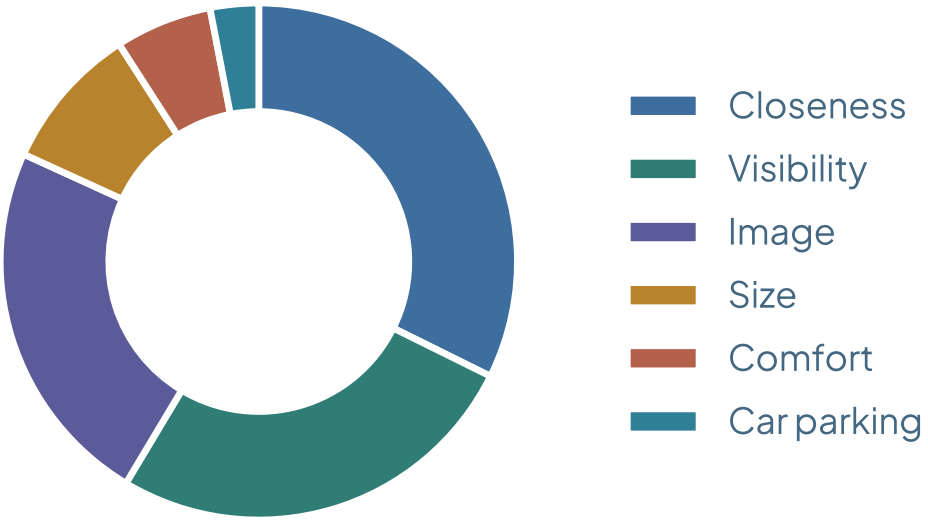
1 The SMART steps

Steps: decision makers, alternatives, value tree, 0–100 values, swing weights (normalised), weighted aggregate, provisional decision, sensitivity. Keep costs separate; drop dominated options; the remaining ones form the efficient frontier; judge whether extra benefit is worth extra cost.

2 Value tree and weights for the office Deep dive



Normalised weights



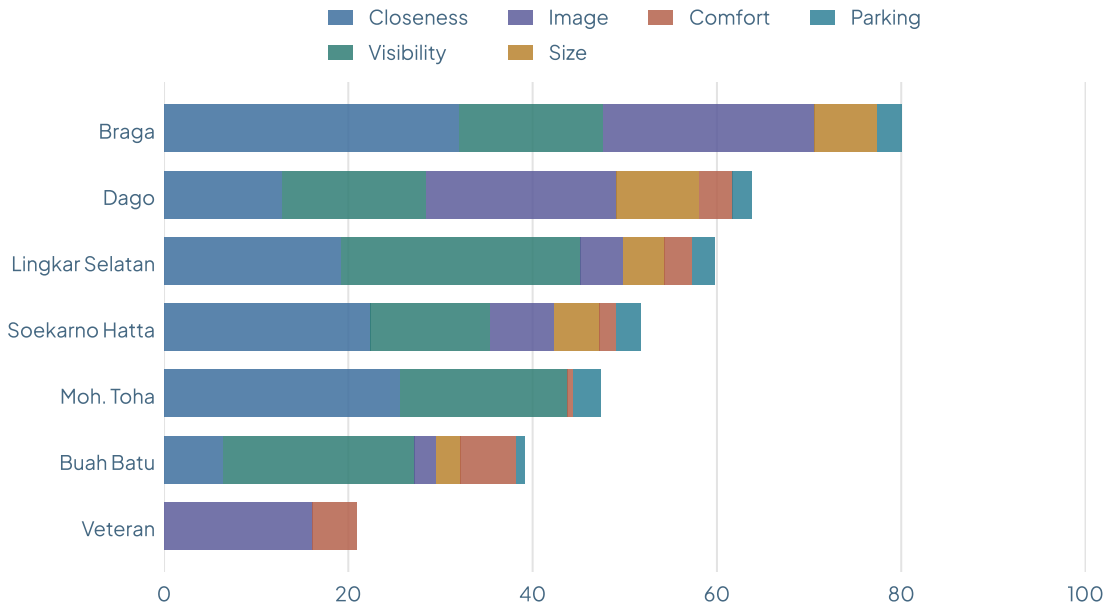
! **Swing weights**
reflect the range between options, not abstract importance.

3 Class example: Bandung office location

Weights: closeness .32, visibility .26, image .23, size .09, comfort .06, parking .03. Braga 80.05 leads, Dago 63.8. Sensitivity: with turnover weight 0, Dago wins (81.49); with working-condition weight 0, Braga rises to 87.2. The real debate is the turnover vs working-condition weighting.

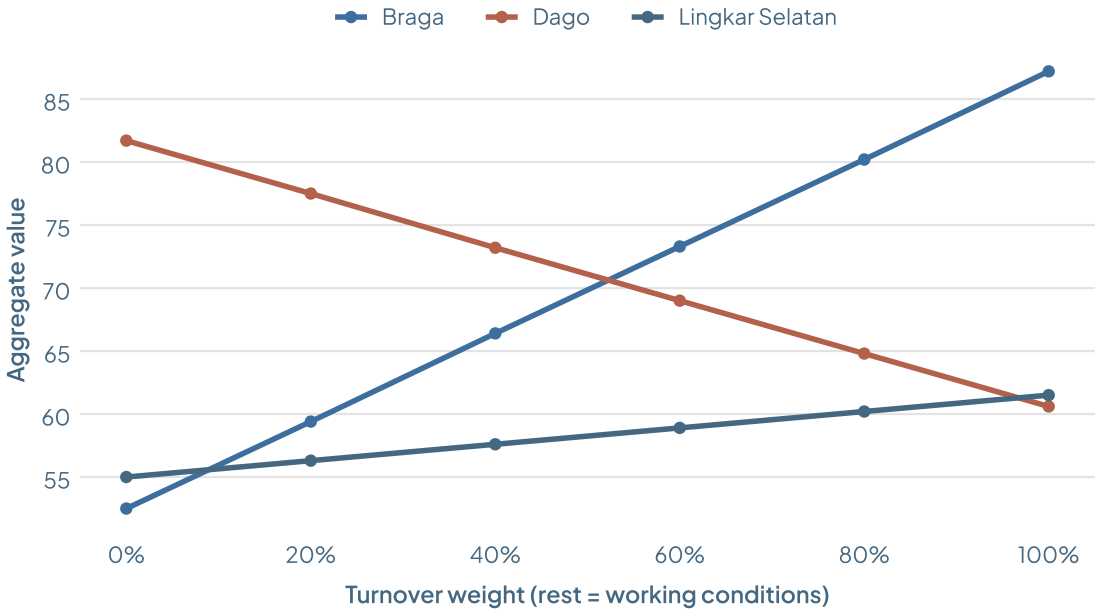
4 Bandung in charts: where the scores come from Deep dive

Contribution of each attribute



Braga wins on closeness and image.

Sensitivity: turnover group weight



Braga and Dago cross at ~52% turnover weight (current 81%).

5 Value functions with the bisection method, and checking SMART results

For numerical attributes (rent, distance, time), 0–100 values should not just follow a straight line. Decision makers often feel improvements non-linearly. A simple way to draw the curve is **bisection**.

≡ **Example: annual office rent**
Worst IDR 100 million = 0, best IDR 40 million = 100.
Question 1: "At what amount X does the improvement from IDR 100m to X feel *as big* as the improvement from X to IDR 40m?" Answer: IDR 75m → value 50.
Question 2: the midpoint between IDR 100m and 75m → IDR 88m → value 25.
Question 3: the midpoint between IDR 75m and 40m → IDR 58m → value 75.
Result: IDR 100m = 0, 88 = 25, 75 = 50, 58 = 75, 40 = 100. The first IDR 25m of savings is "worth" half the scale, and the next IDR 35m is also half — so early savings feel more valuable (a non-linear function). A location with IDR 70m rent scores about 57, not 50 as on a straight line.

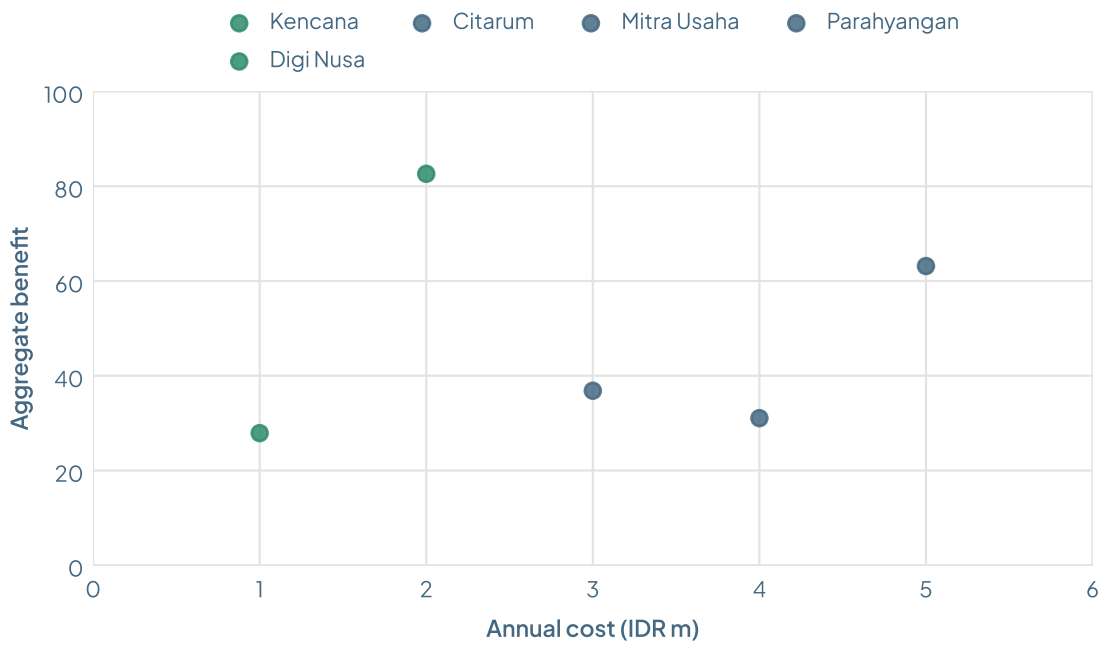
Checking SMART results (Goodwin & Wright call the aim a *requisite* model — sufficient to resolve the problem):

- **Check preference independence.** SMART adds up weighted scores, so attributes must be judged on their own. If the value of "office size" depends on "location" (a big office is only useful in certain locations), the additive model can mislead — merge or split the attributes.
- **Compare with holistic intuition.** If the model's result is very different from the decision maker's feeling, do not simply pick one. Look for the reason: a forgotten attribute? Weights misunderstood (swing vs "importance")? Or is the intuition biased?
- **Sensitivity analysis.** Change the most disputed weights and find the switch point (e.g. Braga and Dago swap places at a turnover weight of about 52%). If the switch point is far from the current weight, the decision is *robust*.
- **Dominance.** An alternative beaten on every attribute by another can be dropped immediately, before computing weights.

6 BUMDes: efficient frontier and RM-weight sensitivity

Deep dive

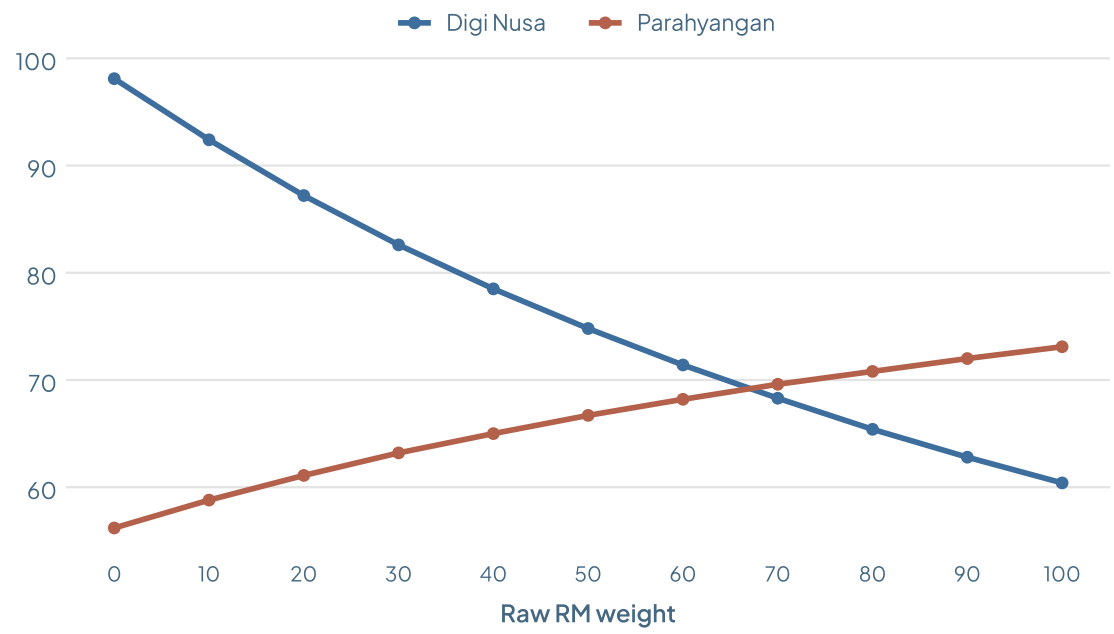
Benefit vs annual cost (BUMDes)



Green = frontier; grey = dominated.

Only Kencana (cheapest) and Digi Nusa remain on the frontier; the others are dominated.

Digi Nusa vs Parahyangan as RM weight changes



Crossover at 67 (≈29.5%).

Key takeaways

1. Value tree, 0–100, swing weights, aggregate, sensitivity.
2. Normalise weights.
3. Separate cost; efficient frontier.
4. Find weight crossovers.
5. Compare cost per benefit point with its value.

Case studies

Case & worked answers

Your syndicate (Week 5): BUMDes Bumi Lestari's bank

Baseline weights (normalised .53 credit line, .26 cash management, .16 RM, .05 branch) give Digi Nusa 82.63 vs Parahyangan 63.16. Sensitivity on RM weight:

crossover at raw weight 67 (~29.5%). Cost–benefit: Parahyangan's extra 12.7 points cost IDR 236,220 per point vs an internal value of ~IDR 13,846 — not justified. Recommendation: Bank Digi Nusa, a robust choice on the efficient frontier.

Worked examples

Problem 1 · Medium **SMART dari awal sampai akhir**

Memilih vendor pelatihan. Swing weight: Kualitas materi 100, Pengalaman trainer 60, Fleksibilitas jadwal 40. Nilai (0–100):

Vendor	Kualitas	Trainer	Jadwal	Biaya
A	100	20	50	Rp15 jt
B	60	100	0	Rp10 jt
C	0	70	100	Rp8 jt

Hitung nilai agregat, cek dominasi, dan harga per poin manfaat antara A dan B.

1. Normalisasi bobot: total 200 → Kualitas 0,5; Trainer 0,3; Jadwal 0,2.
2. A = 0,5×100 + 0,3×20 + 0,2×50 = 50 + 6 + 10 = **66**.
3. B = 0,5×60 + 0,3×100 + 0,2×0 = 30 + 30 + 0 = **60**.
4. C = 0 + 0,3×70 + 0,2×100 = 21 + 20 = **41**.
5. Dominasi: makin mahal makin tinggi manfaatnya (A > B > C), jadi tidak ada yang terdominasi — ketiganya di efficient frontier.
6. A vs B: +6 poin untuk +Rp5 juta → Rp0,83 juta per poin. Layak atau tidak tergantung berapa manajemen menghargai 1 poin.

Answer: A 66, B 60, C 41. Tidak ada yang terdominasi; A hanya layak bila 1 poin manfaat bernilai ≥ Rp0,83 juta.

Practice questions

Try answering first. The answer key is below each question.

1. In SMART the best option on an attribute scores…

A. 1
B. 10
C. 100
D. Its weight

Answer: C. Best 100, worst 0.

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2. Braga's aggregate is…

A. 63.8
B. 75.0
C. 80.05
D. 87.2

Answer: C. Sum of products.

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3. A costlier and less beneficial option is…

A. Efficient
B. Dominated
C. Swing
D. Robust

Answer: B. Drop it.

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4. Digi Nusa stays best while RM weight is below about…

A. 5%
B. 16%
C. 29.5%
D. 53%

Answer: C. Raw weight 67.

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5. Why keep cost separate from benefits?

A. To see IDR per benefit point
B. Cost doesn't matter
C. Higher scores
D. AHP rule

Answer: A. Then judge if extra benefit is worth it.

Sources

- [Lecturer] 04 Decision Analysis: SMART (MK5)
- Goodwin & Wright, Decision Analysis for Management Judgment, Bab 3
- Pre Assignment 5 for Class E, Syndicate 3 — BUMDes Bank Selection (tugas kelompokmu)